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Using Fe K-line emission to separate X-ray binary and millisecond pulsar candidates from cataclysmic variables in the Galactic bulge

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ABSTRACT

Fermi observations have revealed excess γ-ray emission close to the Galactic Center, which could be produced by dark matter annihilation or by unresolved millisecond pulsars (MSPs). We use archival *XMM Newton* and *Chandra* observations of the Galactic Bulge to search for possible MSPs, with the goal of eventually testing the MSP origin hypothesis for the γ-ray excess. We compare the X-rays in the 6.2-7.2 keV energy range with emission in neighboring energy ranges, 5.8–6.2 keV and 7.2–7.6 keV, to search for fluorescent Fe and Fe-K lines in X-ray spectra. We thus create an X-ray color to distinguish continuum sources from line-dominated sources, like cataclysmic variables (CVs). We find 54 *XMM Newton* sources and 83 *Chandra* sources that do not show evidence (either photometric, or via spectral fitting) for Fe lines, among 185 *XMM-Newton* sources & 288 *Chandra* sources with ≥ 250 counts allowing strong tests for lines. These sources lacking Fe lines include known X-ray binaries, pulsar wind nebulae and a magnetar. Thirteen sources lacking Fe lines of unknown nature appear to have radio counterparts. These sources without evidence for Fe lines likely contain many X-ray binaries, millisecond pulsars, some background AGN (~ 34 *XMM-Newton* sources and ~ 5 *Chandra* sources), and possible CVs with very faint Fe emission lines (~ 15 *XMM-Newton* sources and ~ 17 *Chandra* sources). We suggest that detailed follow-up of these unidentified X-ray sources lacking line may uncover the brightest of the expected population of bulge MSPs.

Keywords: Cataclysmic variable stars(203) — X-ray astronomy(1810) — X-ray binary stars(1811) — Millisecond pulsars(1062)

1. INTRODUCTION

The inner kiloparsec of the Milky Way galaxy shows a high density and spherical distribution of stars known as the Galactic Bulge (GB, E. Dwek et al. 1995; C. Wegg & O. Gerhard 2013). With a mass of $\approx 2 \times 10^{10} M_{\odot}$ (E. Valenti et al. 2016; M. Portail et al. 2017) and an age of 7.5–11 Gyr (F. Surot et al. 2019), the GB is an exciting site to study the structure and evolution of the galaxy, formation and evolution of compact objects (e.g. M. P. Muno et al. 2003, 2009; R. Genzel et al. 2010; P. G. Jonker et al. 2011; Z. Zhu et al. 2018), and probe possible dark matter annihilation at the galactic center (GC; e.g. L. Bergström et al. 1998; M. Regis & P. Ullio 2008).

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Fermi observations of the GC show an excess of γ-ray emission from the inner $\sim 1.25°$ of our galaxy, peaking at 2 GeV (e.g. L. Goodenough & D. Hooper 2009; K. N. Abazajian & M. Kaplinghat 2012; C. Gordon & O. Macías 2013; M. Ajello et al. 2016). Some studies have argued that this galactic center excess (GCE) is produced from dark matter annihilation, based on the γ-ray spectra (D. Hooper & L. Goodenough 2011) and the spatial distribution of the GCE (T. Daylan et al. 2016; M. Di Mauro 2021). However, other studies have maintained that the GCE can be explained, both in spectra and morphology, through known astrophysical sources such as unresolved millisecond pulsars (MSPs) formed in the GB (K. N. Abazajian & M. Kaplinghat 2012; Q. Yuan & B. Zhang 2014; T. D. Brandt & B. Kocsis 2015; O. Macias et al. 2018). Identifying the required population of MSPs in the GB would resolve the issue, but this is difficult in the gamma-ray regime (due to the limited angular resolution of gamma-ray telescopes, P. L.

Gonthier et al. 2018), and in the radio (due to distance and interstellar scattering: J. S. Deneva et al. 2009; J.-P. Macquart & N. Kanekar 2015; F. Calore et al. 2016; S. D. Hyman et al. 2019; D. A. Frail et al. 2024).

Most MSPs produce principally blackbody-like X-ray emission (e.g. V. E. Zavlin et al. 2002; S. Bogdanov et al. 2006), that is too faint and soft to be detectable at the distance and extinction typical of the Galactic Center region, as X-ray photons with energies $\lesssim 2$ keV are heavily obscured by the large absorption column towards the GC. However, some MSPs produce relatively bright, magnetospheric, highly pulsed X-ray emission, such as PSR B1821-24 in M28 at $L_X \sim 10^{33}$ erg/s (Y. Saito et al. 1997; W. Becker et al. 2003; R. E. Rutledge et al. 2004; S. Bogdanov et al. 2011), and the non-cluster pulsars PSR J0218+4232 and PSR B1937+21 at $L_X \sim 4 \times 10^{32}$ erg/s (F. Verbunt et al. 1996; L. Kuiper et al. 1998; M. Takahashi et al. 2001; E. V. Gotthelf & S. Bogdanov 2017). Another fraction of MSPs (the "spiders"; M. S. E. Roberts 2013) are in compact binary systems with a non-degenerate companion that produces a wind, generating an intra-binary shock (e.g. Z. Wadiasingh et al. 2018; D. Kandel et al. 2019) which emits bright X-rays with energies $\gtrsim 2$ keV. A significant fraction of redback MSPs, in the field and in globular clusters, are now known to emit hard X-rays with $L_X > 10^{32}$ erg/s (J. Berteaud et al. 2021; J. Zhao & C. O. Heinke 2022; K. I. I. Koljonen & M. Linares 2023). Resolving bright MSPs through X-ray telescopes like *Chandra* and *XMM-Newton* telescopes, which have a better angular resolution than Fermi, will allow us to constrain the MSP population in the GB and probe the source of the GCE.

However, other sources like cataclysmic variables (CVs, e.g, M. Revnivtsev et al. 2009; K. Perez et al. 2015; C. J. Hailey et al. 2016; X.-j. Xu et al. 2019), X-ray binaries (XRBs, e.g., M. Revnivtsev et al. 2008; P. G. Jonker et al. 2011), and background active galactic nuclei (AGN) also emit X-rays with energy $\gtrsim 2$ keV and are detected in X-ray surveys. CVs, some active XRBs, and many nearby AGN show prominent Fe lines — 6.4 keV fluorescent Fe emission, 6.7 keV Fe XXV (He like) K line, and 7.0 keV Fe XXVI (H like) K line (e.g., X.-j. Xu et al. 2016; J. M. Torrejón et al. 2010; A. C. Fabian et al. 2000; C. Ricci et al. 2014). The large amounts of interstellar dust in the Galaxy absorb all optical and ultraviolet information, and the high stellar density towards the GC makes it challenging to identify infrared counterparts of GB sources (C. DeWitt et al. 2010, 2013; S. Greiss et al. 2014; J. Wu et al. 2015). While the detailed spectroscopic analysis of all X-ray sources in the GB would be difficult and time-consuming, selecting a narrower subset of sources based on photometric methods (like broadband hardness ratios (HR) or narrow-band color diagnostics) and analyzing only these sources further would be more efficient.

C. J. Hailey et al. (2018) used the HR between 2-4 and 4-8 keV (we shall call this $HR2$) to select a group of soft non-thermal sources with $HR2 < 0.3$ within a parsec of the Galactic Center, which they argue are X-ray binaries with likely black hole accretors (K. Mori et al. 2021). They distinguish these sources from those with $HR2 > 0.3$, dominated by Fe emission lines, that they argue are dominated by CVs. However, such a broadband hardness ratio can be influenced by Galactic absorption and intrinsic hardness of the X-ray continuum, in addition to the strength of Fe lines.

In this paper, we use X-ray colors, defined narrowly around the Fe-line complex, to select a small set of possible continuum sources. We study these sources in detail as an efficient way to search for candidate MSPs in the GB, along with other continuum sources (X-ray binaries, slower pulsars, and distant AGN). We download the X-ray spectra of GB sources from *Chandra* and *XMM-Newton* data archives and check for any excess Fe emission compared to the continuum spectra of MSPs. We discuss our methodology in detail in §2. We report our results in §3 and discuss the candidate sources in §4. We summarize and discuss the future prospects of our work in §5.

2. DATA ACQUISITION AND ANALYSIS

We use the *XMM-Newton* and *Chandra* observations around the GC to search for potential millisecond pulsar (MSP) candidates. The *XMM-Newton* source catalog (4XMM-DR13, N. A. Webb et al. 2020) and the *Chandra* source catalog (CSC 2.0, I. N. Evans et al. 2010, 2020) provide properties of the sources detected in this region and allow us to access the associated data products directly, including the spectra. We use these spectra for our analysis.

2.1. *XMM-Newton sources*

The X-ray telescope *XMM-Newton* has extensively observed the region around the GC. G. Ponti et al. (2015) presents exposure maps reaching to 1.5 Ms equivalent of EPIC-pn observations around the central 1.5° from the GC. For our purpose, we select a 6°× 6° field around the GC. This region is characterized by high interstellar extinction and substantial absorption column density. The absorption column density also varies significantly within this region, particularly affecting soft X-rays. We only consider regions with $N_H > 5 \times 10^{22}$ cm^{-2}, to identify only highly absorbed sources that are consistent

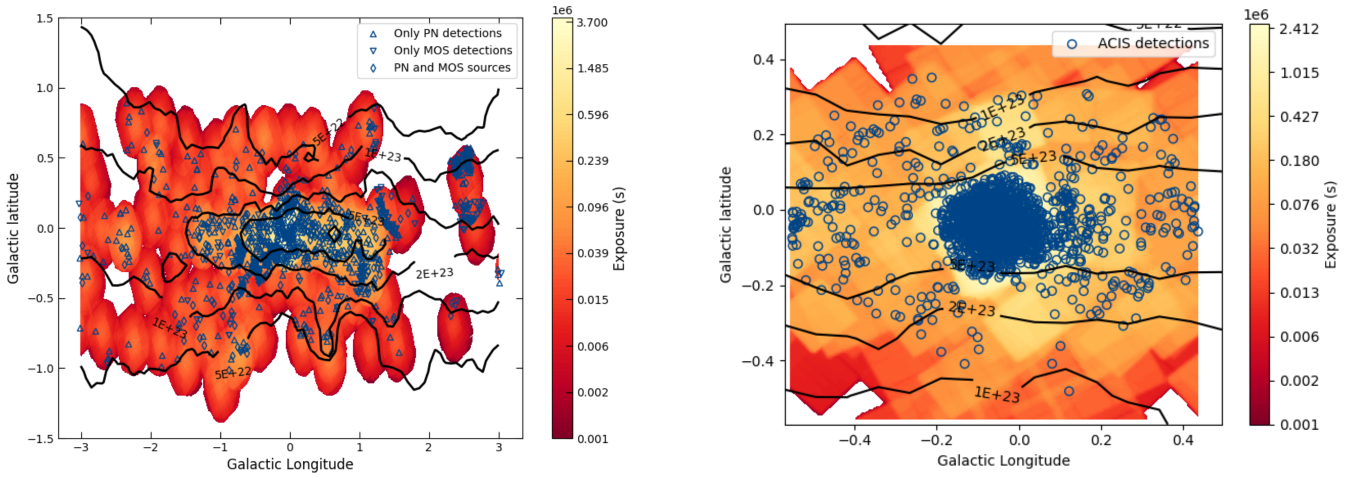

Figure 1. *XMM-Newton* EPIC (*left*) and *Chandra* ACIS (*right*) observations around the Galactic center. We combine the EPIC-PN and EPIC MOS exposures, using a weight of 0.4 for MOS data to account for the differing effective areas. We also overlay contours of N_H, and positions of all the sources we consider. Sources detected in EPIC-PN but not EPIC-MOS (blue upper triangle), detected in EPIC-MOS but not EPIC-PN (downward blue triangle), and detected in both EPIC-PN and MOS (blue diamond) are marked separately. *Chandra* ACIS sources are shown in blue circles. We only consider *XMM Newton* sources with $N_H \geq 5 \times 10^{22}$ cm^{-2} and *Chandra* sources within $30'$ of the Galactic Center.

with GC distances (e.g. M. P. Muno et al. 2003). We model the interstellar extinction from the Galactic Dust Reddening and Extinction Service (D. J. Schlegel et al. 1998; E. F. Schlafly & D. P. Finkbeiner 2011) and use the correlation reported in A. Bahramian et al. (2015) to calculate the N_H from extinction values (i.e. we use regions where $A_v >= 17.8$).

We then look into the 4XMM-DR13 catalog (N. A. Webb et al. 2020) to search for sources in this region and the corresponding *XMM Newton* EPIC observations. We only select point sources (`EP_EXTENT = 0`), and require at least one spectrum (`SPECTRA=True`). The GC region has bright diffuse emission and is crowded with many bright sources. This leads to several moderately bright sources where the summary flag of the detection, `SUM_FLAG`, is flagged as 1 (i.e. source parameters may be affected). **In this paper, we directly use the extracted spectra in the 4XMM-DR13 catalog for detailed analysis, and do not use any of the extracted source parameters (other than hardness ratio HR$_3$ for selecting sources in the GB). Manual inspection of a few detections flagged as "1" does not show any obvious errors in the extraction regions of the source and background spectra.** We consider detections flagged as 0 or 1 for our analysis, and ignore detections flagged > 1. We use the hardness ratio, HR$_3$, provided in the source catalog to remove likely foreground sources. This hardness ratio is calculated from count rates (CRs) in bands 3 (1–2 keV) and 4 (2–4.5 keV) i.e. $HR_3 = (CR_4 - CR_3)/(CR_4 + CR_3)$. Given the high N_H in our regions of interest, we expect very few photons with energy < 2 keV. We therefore only choose sources with 1σ upper limits on $HR_3 > 0.65$ for our analysis (consistent with XMM observations of a source with $N_H = 5 \times 10^{22}$ cm^{-2} and $\Gamma = 1.5$). This gives us **726** X-ray sources, with 2147 detections in total **(673 of these detections have `SUM_FLAG` = 1; 610 sources have at least 1 detection with `SUM_FLAG` = 0, 257 sources have at least 1 detection with `SUM_FLAG` = 1)**. We extract the EPIC-PN and EPIC-MOS spectra of these **726** X-ray sources from the HEASARC data archive. We combine the spectra for sources detected across multiple observations using the `epicspeccombine` method. We process EPIC-PN and EPIC-MOS spectra separately as their responses are very different. We show the exposure map, absorption column contours around the Galactic Center (D. J. Schlegel et al. 1998), and the selected XMM sources in Fig. 1.

For comparison, we also simulate 20000 spectra of highly absorbed MSPs, as they would be observed by XMM-Newton EPIC-PN and EPIC-MOS detectors. Since we do not know the exact distribution of the MSP properties in the GB, we use a log-uniform distribution for absorption column density, N_H, between 5×10^{22}cm^{-2} and 5×10^{23} cm^{-2}; a uniform distribution of power-law indices, Γ, between 1.0 and 2.0; and a log-uniform distribution of 2–10 keV X-ray luminosity L_X between 10^{31} ergs s^{-1} and 10^{34} ergs s^{-1}. The lower limit on N_H is consistent with our selection of sources and the higher limit is where $\gtrsim 50\%$ of the flux is absorbed in the 6–7 keV range. The limits on Γ are based on the distribution among the bright MSPs detected in GCs (J. Zhao & C. O. Heinke 2022). For each simulated MSP spectrum, we use the combined exposure, background, and response of a randomly selected XMM observation; values of N_H, Γ, and L_X chosen at random from their corresponding distributions; and then generate the fake spectrum through the `fakeit` method in XSPEC v12.9. If an observed source is not detected in one of the detectors, the MSP spectrum for that detector is not generated.

2.2. *Chandra sources*

The *Chandra* X-ray observatory has ≈ 2 Ms of total ACIS-I observations pointed towards Sgr A*. These are ideal for probing sources close to Sgr A*, due to the depth of these observations and *Chandra's* excellent spatial resolution in this crowded (and relatively high-background) environment. Farther from Sgr A*, XMM's angular resolution is generally sufficient, and as XMM collects more photons per second, XMM detections often deliver superior spectra. Thus, we only use the central $30'$ around Sgr A* for analysis of *Chandra* data. We only consider sources and regions with $N_H \geq 5 \times 10^{22}$, to be consistent with the Galactic Center distance (consistent with XMM constraints). We only use ACIS observations without a grating inserted (cf. M. A. Nowak et al. 2012; Z. Zhu et al. 2018), to allow for combining spectra from multiple observation epochs.

We use the Chandra source catalog (CSC) v2.0 (I. N. Evans et al. 2010, 2024) to search for sources in this region. We select sources that have a `significance` greater than 3σ and `likelihood_class = TRUE` (`likelihood_class` is based on false detection rate, and not solely on the source significance or likelihood of detection). We require that the best estimate of the net source flux and photon flux in the "hard" (2–7 keV) band (`flux_aper_h`, `photflux_aper_h`) is positive, and that the source has at least one detection stack where the significance of the detection in the "hard" band (`s.detect_significance_h`, `s.flux_significance_h`) is higher than 3σ. We also set the upper limit of the hardness ratio between the "hard" and "medium" (1.2–2 keV) bands, $HR_{hm,upper} > 0.7$ to remove foreground sources. CSC v2.0 does not specify HR_{hm} or its error limits for sources if its aperture flux values and errors cannot be determined in either the "hard" or the "medium" bands. Since several highly absorbed sources may not be detected below 2 keV, i.e. in the "medium" band, we also allow for sources where HR_{hm} is not specified (we have already selected that `photflux_aper_h > 0`, so this only adds hard sources with non-detection in the "medium" band). Manually inspecting the CSC catalog shows that the confusion flag, `conf_flag`, and extended source flag, `extent_flag`, are not very accurate for sources close to the GC (due to the bright background diffuse emission). Therefore we do not constrain these properties. We also require that the "hard" band flux significance in the selected observations (`o.flux_significance_h`) is greater than 3σ, and that the off-axis angle is less than $8.0'$. This gives us 2068 sources with 34761 observations. We extract spectra and detector responses of the selected sources in the selected observations from the Chandra Data Archive. While the *Chandra* ACIS soft energy response has degraded significantly over the years[3], its 2–10 keV response has remained nearly constant. Therefore, we combine the source spectra from all the observations using the `combine_spectra` tool in CIAO. We also generate 10000 simulated *Chandra* MSP spectra with the same N_H, Γ, and L_X used for XMM spectra, and the combined background and responses observed for the detected *Chandra* ACIS sources.

The properties of *Chandra* sources within the central parsec may be particularly affected by bright diffuse emission, which can be tricky to address with automated extraction routines (cf. C. J. Hailey et al. 2018).

2.3. *Simulated source selection*

The GB has large amounts of diffuse X-ray emission, leading to a high background for the simulated and detected sources. Simulating sources by randomly choosing N_H, L_X, and background can lead to some low-luminosity sources being associated with high N_H and/or background; such sources would not be detected in observations. We only consider simulated spectra with more than 3σ (net counts by square root of background counts) significance of detection in the 2–10 keV range for analysis. Our simulations will also include bright sources simulated with low backgrounds. However, these do not significantly affect our results as their spectra are dominated by the Poisson noise from source emission rather than the background. Manually inspecting the source and background extraction regions in the 4XMM-DR13 shows that the selected background regions for some of the sources may be incorrect and lead to over-subtraction of the background spectra. Limiting ourselves to observed spectra with more than 3σ sources allows us to remove such sources.

Fig. 2 compares the mean normalized X-ray spectra of observed and simulated *XMM-Newton* and *Chandra*

[3] https://cxc.cfa.harvard.edu/proposer/POG/html/chap6.html

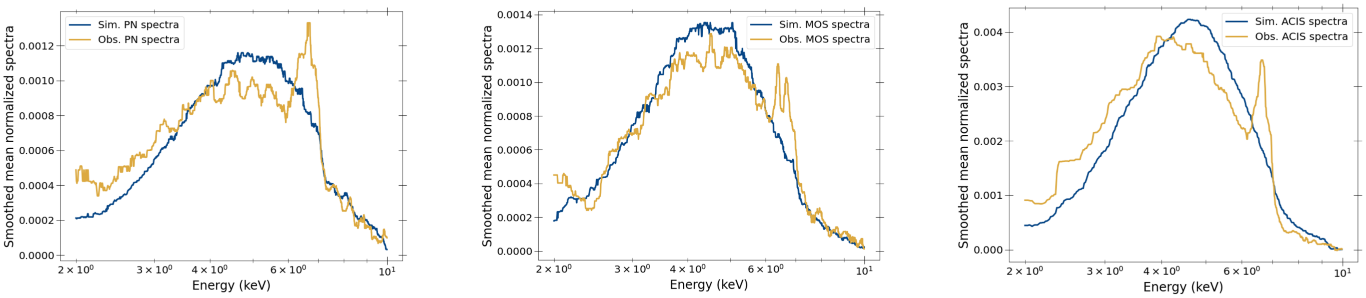

Figure 2. Mean normalized spectra of simulated MSPs and observed sources with EPIC-PN (*left*), EPIC-MOS (*center*), and ACIS (*right*) detectors. Each spectrum is divided by its net counts in the 2–10 keV energy range to get its normalized spectrum. The mean of the normalized spectra for observed sources shows an excess in 6–7 keV corresponding to the fluorescent Fe (6.4 keV) and Fe-K (6.7 keV & 7.0 keV) lines. This emission is from CVs embedded in our sample. The observed spectra also differ from simulated spectra in terms of hardness (possibly due to a more complex N_H & L_X distribution) and apparent absorption features (due to over-subtraction of the background).

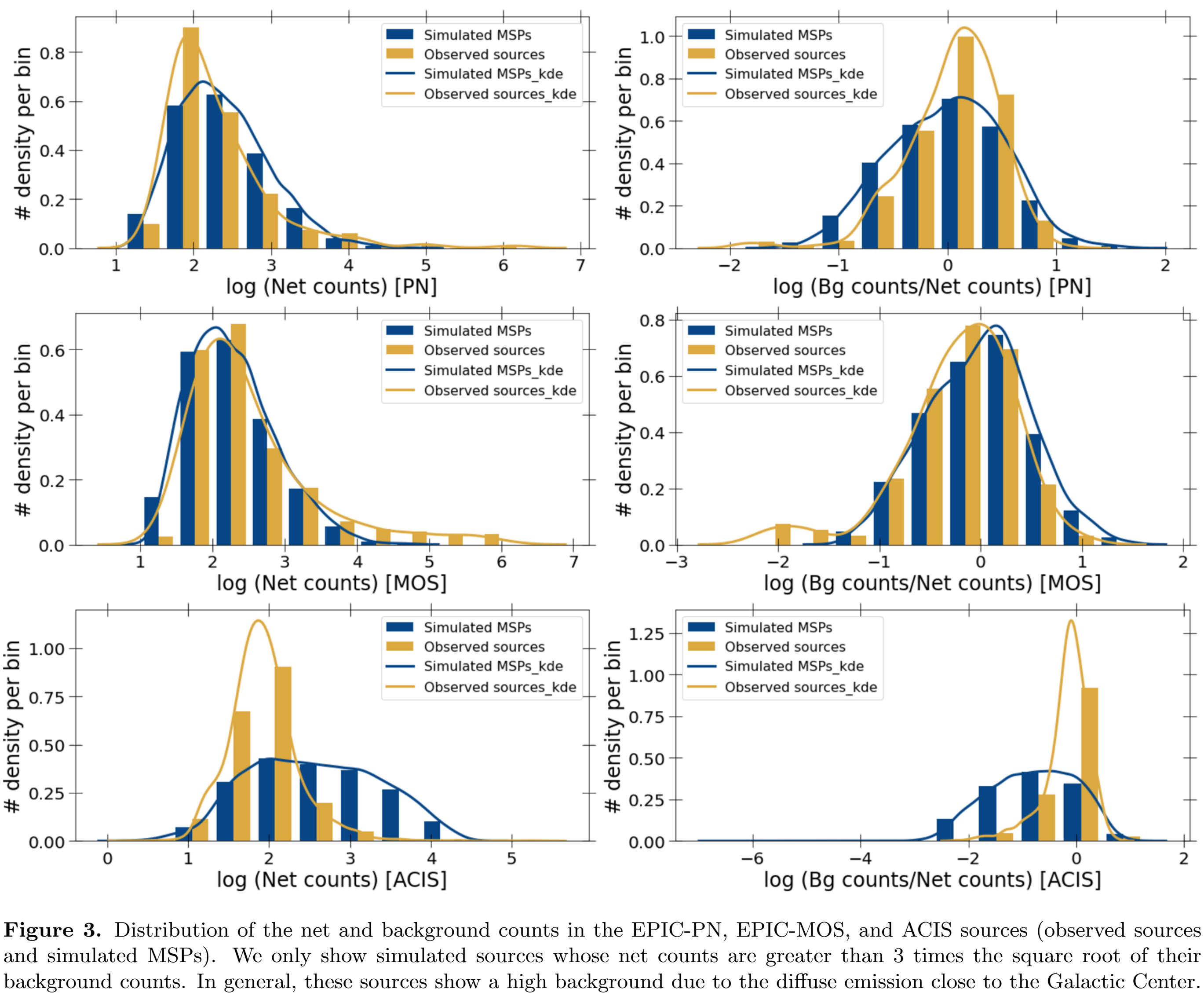

Figure 3. Distribution of the net and background counts in the EPIC-PN, EPIC-MOS, and ACIS sources (observed sources and simulated MSPs). We only show simulated sources whose net counts are greater than 3 times the square root of their background counts. In general, these sources show a high background due to the diffuse emission close to the Galactic Center. Our simulated sources cover the entire range of net counts and background contributions spanned by the observed sources.

sources. The mean normalized spectrum is calculated by dividing the spectrum of each source by its net counts in the 2–10 keV energy range, and calculating the mean across all sources. Taking the mean spectra after such a normalization allows us to see the features in the faint spectra without being masked by those of bright sources. We see that the mean normalized spectra of the observed sources show excess emission in the 6–7 keV range, when compared with the normalized spectra of simulated MSPs. This excess emission is likely from CVs, X-ray binaries, and background AGN embedded among the observed sources that emit fluorescent Fe and Fe-K lines (6.4, 6.7 keV and 7.0 keV). Fig. 2 also shows that the mean normalized spectra of observed PN and ACIS sources have higher 2-4 keV emission, and the mean normalized spectrum of observed MOS sources shows a dip at ~ 2.5 keV. These features could be due to differences in the inherent distribution of N_H and L_X, or improper background subtraction. Limiting ourselves to the 5.5-7.5 keV energy interval allows a more robust comparison of simulated and observed sources.

We compare the distribution of net counts, background-to-net ratio, and signal-to-noise ratio of simulated and observed sources in EPIC-PN, EPIC-MOS, and ACIS detectors in Fig. 3. While the exact distributions of these properties for the simulated sources and observed sources are different, the simulated sources cover the entire range that the observed source and background counts cover.

2.4. *X-ray color analysis.*

We use the simulated spectra to analyze the distribution of X-ray colors for MSPs (and other quiescent X-ray sources) that do not show any excess Fe emission. We define our X-ray colors as the ratio between the net counts in 6.2–7.2 keV and the sum of net counts in 5.8–6.2 and 7.2–7.6 keV. The 6.2–7.2 keV emission range includes the Fe XXV ($7 \rightarrow 1$) 6.7 keV line and the Fe XXVI ($3 \rightarrow 1$, $4 \rightarrow 1$) 6.95, 6.97 keV lines[4], as well as the 6.4 keV line from neutral Fe. The regions 5.8–6.2 keV and 7.2–7.6 keV do not show any lines for plasma with $kT > 5$ keV. So, we choose them as the continuum regions with which the 6.2–7.2 keV emission can be compared.

We plot the X-ray colors of the simulated sources against the net counts in the 2–10 keV energy range to constrain the X-ray color distribution of MSPs. We divide the simulated MSPs into 20 bins based on the logarithm of 2–10 keV net counts. **For each bin of the net count distribution, we calculate the 95% confidence interval on X-ray colors by identifying the 2.5% & 97.5% percentiles as upper and lower limits of the MSP colors. We use these bounds to identify candidate sources lacking iron lines.**

3. RESULTS

Fig. 4 compares the distribution of X-ray color and net counts in the simulated MSPs and the observed sources, across EPIC-PN, EPIC-MOS, and ACIS detectors. As expected, the scatter in the X-ray color decreases as the net counts of the simulated MSPs increase (i.e. the signal-to-noise ratio increases). To define a 2-sigma range in X-ray color for each net counts bin of our simulated sources (left panels of Fig. 4), we assume a Gaussian distribution of the X-ray color. Note that our definition of an X-ray color is a ratio of two Poisson variables, resulting in a heavier tail than a strict Gaussian distribution. Therefore we calculate the standard deviation using 5σ-clipping; i.e., we exclude values $> 5\sigma$ away from the mean, and calculate σ iteratively, until convergence.

The scatter in the observed source colors (right panels of Fig. 4) is higher than that of simulated MSPs even at high net counts, specifically towards higher X-ray colors. This is due to sources with fluorescent Fe and Fe-K lines (i.e. CVs, XRBs and AGN) in the observed sample. Fig. 4 shows the 2σ X-ray color limits for the simulated sources as a pair of black lines. We expect the observed MSPs to mostly lie within these bounds. We only consider EPIC-PN, EPIC-MOS, and ACIS sources with net counts ≥ 250, as the 2σ bounds on the X-ray colors get very large for sources with low net counts. These two constraints give us **104** EPIC-PN, **70** EPIC-MOS (**45** in common between PN and MOS selections), and **144** Chandra ACIS candidate continuum sources. Differences in the selected set of PN and MOS sources could be due to our imposed counts cutoff, to sources falling on the corners of chips or in chip gaps, to scattering from source or background noise, etc.

One concern is whether variation in N_H among real sources could artificially blur the distribution. We included a wide variation in N_H (5×10^{22} to 5×10^{23} cm^{-2}) in our simulated sources (§2.1), but high-count simulated sources have well-determined Fe colors (see Fig. 4, left). Fig. 4, right, also shows that the distribution of X-ray colors for bright X-ray sources (largely X-ray binaries) is tight and centered around small EWs, appropriate for continuum sources, and that the predicted lower bound on X-ray colors from the simulations matches the observed lower bound to X-ray colors at all net count values.

[4] See, e.g., http://app.atomdb.org

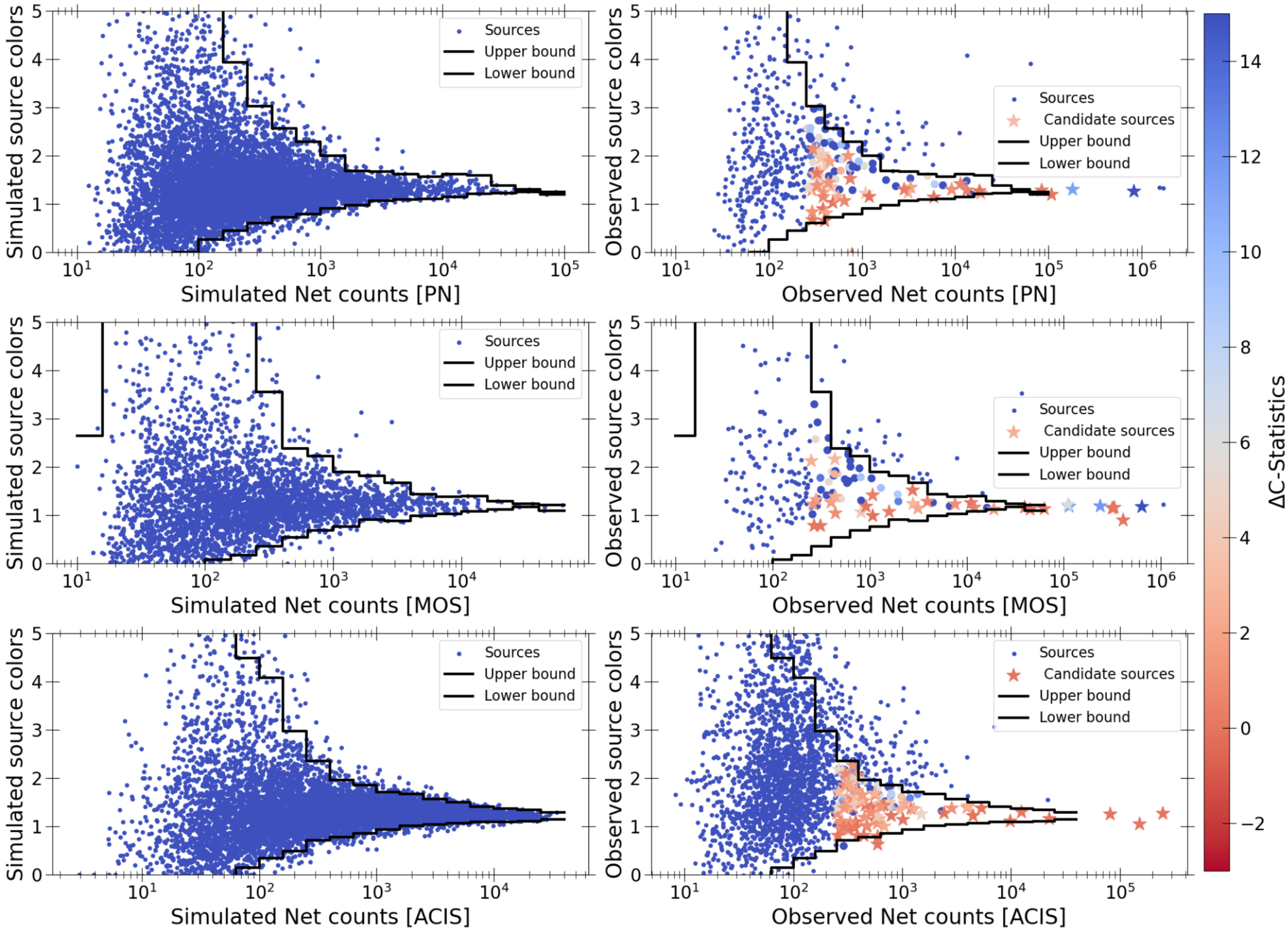

Figure 4. Distribution of simulated MSPs and observed sources in the X-ray color vs. net counts plane. We calculate the X-ray color as the ratio of the net counts in the 6.2–7.2 keV region, to the sum of the net counts in the 5.8–6.2 and 7.2–7.6 keV regions, so larger colors indicate Fe line emission. We also show the 2σ bounds on the color calculated empirically from our simulated sample of MSPs. The observed sources show a greater scatter in the X-ray colors, particularly to higher X-ray color, compared to the simulated MSPs, due to the presence of line-emitting CVs among the observed sources. Our final list of candidate continuum sources, that appear to lack lines both by X-ray color and in spectral fitting, are marked as stars.

We then perform spectroscopic analysis on this selected sample of sources, using XSPEC v12.11.1. We compare three models – an absorbed power-law (model I), an absorbed power-law with a Gaussian (Fe) line, with a fixed width of 0.1 keV and energy free to vary from 6.3 to 7.1 keV (model II), and an absorbed power-law with three Gaussian lines with fixed positions at 6.4, 6.7 and 7.0 keV, and 0 keV width (model III). We assume J. Wilms et al. (2000) interstellar abundances and perform the spectral fitting in the 2–10 keV energy range using Cash statistics (C-statistics). Model II has 2 more free parameters than model I, while model III has 3 more free parameters than model I. We show the difference in the C-statstics between model I and model III fits for the selected sources in Fig. 4. We summarize the results of our spectroscopic analysis of these selected source in the Table 2. The full version of the table will be available online. We describe the columns of the table here.

For a source to be identified as a continuum source, i.e. that model I is more likely than model II or model III, we require that either a) ($cstat_I < cstat_{II} + 4$) & ($cstat_I < cstat_{III} + 6$); or b), that the normalization of all the Gaussian lines in models II and III are consistent with zero (i.e., the lower limit of the 90% confidence error bars reaches zero). Here, $cstat_I$, $cstat_{II}$, and $cstat_{III}$ are the best fitting C-statistic values for models I, II and III respectively. Criterion a) essentially applies the Akaike Information Criterion, by comparing the C-statistic, a log likelihood, of the models, applying a penalty that increases with the number of free parameters, and criterion b) directly checks for the log-likelihood of the Fe emission line(s). Using these constraints, we find 46 continuum EPIC-PN sources, **33** continuum EPIC-MOS sources (55 continuum XMM sources in total), and 83 ACIS continuum sources.

The continuum emission of an X-ray source might be more complicated than a simple power-law. The X-ray spectra can be further complicated by sources falling on chip gaps in some observations, improper selection of the background, etc. Therefore, the quality (or "goodness") of the absorbed power-law fit alone is not a good indicator of a continuum source, and we do not use it as a criterion. These complications in the X-ray spectra are more evident when fitting a bright source; e.g., models II and III can sometimes fit the spectra better with spurious Gaussians that do not coincide with the expected positions and equivalent widths of Fe lines. Therefore we include c) all sources with $> 10^5$ counts with colors consistent with simulated MSPs in our list of line-less sources. These are mostly bright X-ray binaries. The positional distribution of the candidate sources is shown in Fig. 5.

Regarding the issue of difficult extraction of sources in the central parsec, we note that only 12 of the 83 candidate MSP *Chandra* sources lie in the central parsec. Thus, even if confusion issues were strong there, they would not significantly affect our results as a whole.

4. DISCUSSION

4.1. *X-ray sources with counterparts*

First, we check our candidate continuum source-list against known sources close to the Galactic center. We find 16 known sources in our list - 14 X-ray binaries/transients, one pulsar wind nebula (PWN), and one magnetar (SGR J1745-2900, K. Mori et al. 2013; N. Rea et al. 2013). Checking the spectra of the X-ray binaries shows that these do not show Fe fluorescent lines during their X-ray outbursts. The PWN, 4XMM J174722.8-280904, in the supernova remnant G0.9+0.1, was only detected in our *XMM-Newton* data (outside our *Chandra* source selection region), which does not have the resolution to distinguish the PWN from the likely pulsar CXOU J174722.8-280915 (B. M. Gaensler et al. 2001; D. Porquet et al. 2003; F. Camilo et al. 2009).

We also check for other multi-wavelength counterparts of our candidate X-ray sources. Due to the large number of stars and dust clouds in the Galactic plane, the probability of finding optical and infrared counterparts by coincidence is very high. Therefore, we limit ourselves to radio counterparts with $\nu \leq 5.5$ GHz. We start by searching for counterparts within $2''$ of the given source position (a conservatively large error radius for astrometric offsets between catalogs). The probability of random matches is high for radio surveys around the GC. We randomly shift the positions of the candidate X-ray sources by $10''$ and check for the number of chance counterparts. We perform 100 such realizations to estimate the false alarm rates (FAR), i.e. the expected number of XMM and Chandra sources that could have spurious matches given the separation distance between the X-ray and radio source position. We only select radio sources where the separation corresponds to an FAR < 0.2. We find counterparts to eight *Chandra* sources in the Jansky Very Large Array S-band observations of 2014 and 2019 (J.-H. Zhao et al. 2020), one *Chandra* source in the 3 GHz Very Large Array Sky Survey quick-look images (Y. A. Gordon et al. 2021), and one *XMM-Newton* source in the VLA C-band (X. Lu et al. 2019) observations (in addition to known sources). Four of these sources have been suggested to be associated with PWN (S. Park et al. 2005; M. P. Muno et al. 2008; J.-H. Zhao et al. 2013). We summarize these counterparts in Table 3 and Table 4. The full version of the tables are available online. Here we only show the first three candidates from the Chandra observation.

4.2. *X-ray variability*

We also check for X-ray variability of our candidate sources in the 4XMM-DR13 and CSC v2.0 catalogs. Although the 4XMM catalog has a variability flag, it only considers variability within the observation/detection. In some cases, the source variability flag `SC_VAR_FLAG` is set to True (T) due to observed variability in a detection with summary flag, `SUM_FLAG` > 1. In addition, we choose to use a looser threshold for variability than the strict $p < 10^{-5}$ used in the 4XMM-DR13 catalog (p is the probability that the flux is constant). The CSC v2.0 catalog uses a much looser threshold for variability, $p < 0.1$, and could mark a source variable in the master catalog even when some of the source detections in the stack might be confusing (e.g., high off-axis angle, low significance of detection, etc.). Therefore, we use other source and detection properties in these catalogs to determine intra- and inter-observation variability.

The 4XMM-DR13 catalog also gives the chi-square probability that the observed detection is constant, `EP_CHI2PROB`. We check for these values among the "good" detections that we have analysed here (check §2.1 for the selection criteria). For sources with multiple detections we use the minimum value of `EP_CHI2PROB`. We set a threshold probability, $p < 3.3 \times 10^{-4}$ to check if the source is variable on short time scales (i.e., within an observation). Given that we have 54 candidate sources

Table 1. Descriptive version of the "Spectral fitting of *XMM-Newton* sources" table

Number	Column name	Units	Explanation
1	Name	—	IAU name of the XMM source
2	Source_ID	—	Unique source identifier in the 4XMM-DR13 catalogue
3	Counts_PN	photons	2–10 keV net counts in the combined PN spectra
4	Color_PN	—	Fe color in the combined PN spectra
5	Counts_MOS	photons	2–10 keV net counts in the combined MOS1 and MOS2 spectra
6	Color_MOS	—	Fe color in the combined MOS1 and MOS2 spectra
7	NH[1]	cm^{-2}	Best fitting absorption column density for the source
8	NH_low[1]	cm^{-2}	Lower limit of absorption column density
9	NH_high[1]	cm^{-2}	Upper limit of the absorption column density
10	PL_index[1]	—	Best fitting power law index for the source
11	PL_index_low[1]	—	Lower limit of the power-law index
12	Pl_index_high[1]	—	Upper limit of the power-law index
13	Flux[1]	ergs cm^{-2} s^{-1}	2–10 keV flux
14	Flux_low[1]	ergs cm^{-2} s^{-1}	Lower limit of 2–10 keV flux
15	Flux_high[1]	ergs cm^{-2} s^{-1}	Upper limit of 2–10 keV flux
16	C-statistics	—	C-statistics for the best fitting absorbed power-law model
17	DOF	—	Degrees of freedom for the absorbed power-law model
18	Line_pos[2]	keV	Position of the Gaussian line in model II
19	Line_pos_low[2]	keV	Lower limit of the Gaussian line position
20	Line_pos_high[2]	keV	Upper limit of the Gaussian line position
21	Line_photflux[2]	photons cm^{-2} s^{-1}	Photon flux from the Gaussian line
22	Line_photflux_low[2]	photons cm^{-2} s^{-1}	Lower limit of the photon flux from Gaussian line
23	Line_photflux_high[2]	photons cm^{-2} s^{-1}	Upper limit of the photon flux from Gaussian line
24	Delta_cstat_II-I	—	Difference in C-statistics between model I and II
25	Photflux_6.4[3]	photons cm^{-2} s^{-1}	6.4 keV line photon flux
26	PhotFlux_6.4_low[3]	photons cm^{-2} s^{-1}	Lower limit of 6.4 keV line photon flux
27	Photflux_6.4_high[3]	photons cm^{-2} s^{-1}	Upper limit of the 6.4 keV line photon flux
28	Photflux_6.7[3]	photons cm^{-2} s^{-1}	6.7 keV line photon flux
29	Photflux_6.7_low[3]	photons cm^{-2} s^{-1}	Lower limit of 6.7 keV line photon flux
30	Photflux_6.7_high[3]	photons cm^{-2} s^{-1}	Upper limit of 6.7 keV line photon flux
31	Photflux_7.0[3]	photons cm^{-2} s^{-1}	7.0 keV line photon flux
32	Photflux_7.0_low[3]	photons cm^{-2} s^{-1}	Lower limit of 7.0 keV line photon flux
33	Photflux_7.0_high[3]	photons cm^{-2} s^{-1}	Upper limit of 7.0 keV line photon flux
34	Delta_cstat_III-I[3]	—	Difference in C-stat between models I and III

NOTE—This Table is published in its entirety in the electronic edition of the *Astrophysical Journal*. Here, we only describe the columns for guidance regarding its form and content.

[1]Values correspond to model I (`tbabs*pegpw`)

[2]Values correspond to model II (`tbabs*(pegpw+gaussian)`)

[3]Values correspond to model III (`tbabs*(pegpw+gaussian+gaussian+gaussian)`)

Table 2. Descriptive version of the "Spectral fitting of *Chandra* sources" table

Number	Column name	Units	Explanation
1	Name	—	IAU name of the XMM source
2	Counts	photons	2–10 keV net counts in the combined ACIS spectra
3	Colors	—	Fe color in the combined ACIS spectra
4	NH[1]	cm^{-2}	Best fitting absorption column density for the source
5	NH_low[1]	cm^{-2}	Lower limit of absorption column density
6	NH_high[1]	cm^{-2}	Upper limit of the absorption column density
7	PL_index[1]	—	Best fitting power law index for the source
8	PL_index_low[1]	—	Lower limit of the power-law index
9	Pl_index_high[1]	—	Upper limit of the power-law index
10	Flux[1]	ergs cm^{-2} s^{-1}	2–10 keV flux
11	Flux_low[1]	ergs cm^{-2} s^{-1}	Lower limit of 2–10 keV flux
12	Flux_high[1]	ergs cm^{-2} s^{-1}	Upper limit of 2–10 keV flux
13	C-statistics	—	C-statistics for the best fitting absorbed power-law model
14	DOF	—	Degrees of freedom for the absorbed power-law model
15	Line_pos[2]	keV	Position of the Gaussian line in model II
16	Line_pos_low[2]	keV	Lower limit of the Gaussian line position
17	Line_pos_high[2]	keV	Upper limit of the Gaussian line position
18	Line_photflux[2]	photons cm^{-2} s^{-1}	Photon flux from the Gaussian line
19	Line_photflux_low[2]	photons cm^{-2} s^{-1}	Lower limit of the photon flux from Gaussian line
20	Line_photflux_high[2]	photons cm^{-2} s^{-1}	Upper limit of the photon flux from Gaussian line
21	Delta_cstat_II-I	—	Difference in C-statistics between model I and II
22	Photflux_6.4[3]	photons cm^{-2} s^{-1}	6.4 keV line photon flux
23	PhotFlux_6.4_low[3]	photons cm^{-2} s^{-1}	Lower limit of 6.4 keV line photon flux
24	Photflux_6.4_high[3]	photons cm^{-2} s^{-1}	Upper limit of the 6.4 keV line photon flux
25	Photflux_6.7[3]	photons cm^{-2} s^{-1}	6.7 keV line photon flux
26	Photflux_6.7_low[3]	photons cm^{-2} s^{-1}	Lower limit of 6.7 keV line photon flux
27	Photflux_6.7_high[3]	photons cm^{-2} s^{-1}	Upper limit of 6.7 keV line photon flux
28	Photflux_7.0[3]	photons cm^{-2} s^{-1}	7.0 keV line photon flux
29	Photflux_7.0_low[3]	photons cm^{-2} s^{-1}	Lower limit of 7.0 keV line photon flux
30	Photflux_7.0_high[3]	photons cm^{-2} s^{-1}	Upper limit of 7.0 keV line photon flux
31	Delta_cstat_III-I[3]	—	Difference in C-stat between models I and III

NOTE—This Table is published in its entirety in the electronic edition of the *Astrophysical Journal*. Here, we only describe the columns for guidance regarding its form and content.

[1]Values correspond to model I (`tbabs*pegpw`)

[2]Values correspond to model II (`tbabs*(pegpw+gaussian)`)

[3]Values correspond to model III (`tbabs*(pegpw+gaussian+gaussian+gaussian)`)

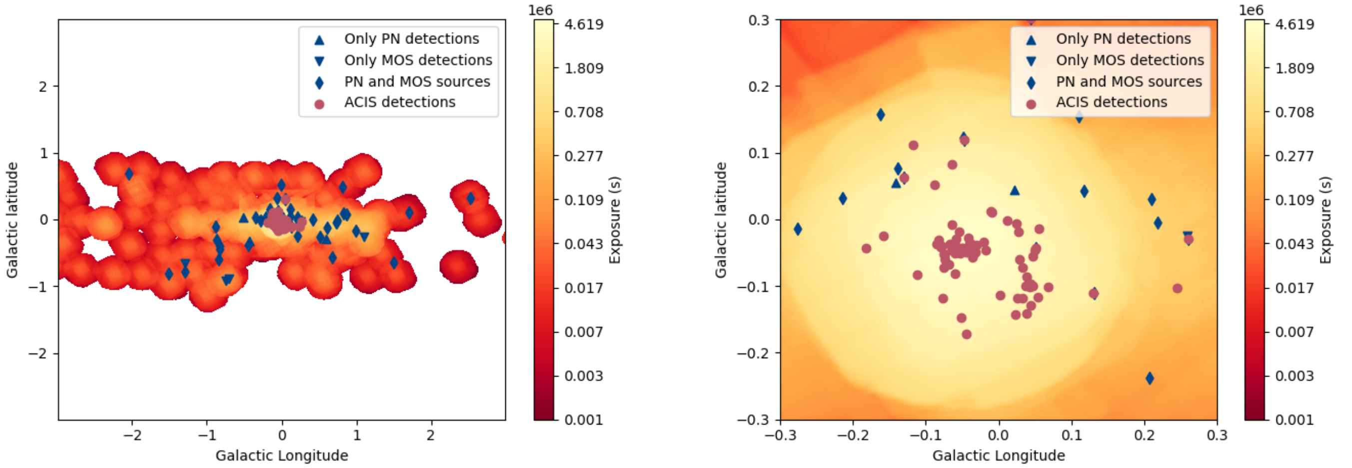

Figure 5. Sky positions of candidate continuum sources detected in *XMM-Newton* (blue circles) and *Chandra* (red circles) observations. Among the XMM detections, sources only identified through EPIC-pn (EPIC-MOS) analysis are shown as upward (downward) triangles, and sources detected in both EPIC-pn and EPIC-MOS are marked as diamonds. *Left*, a view of the full region. *Right*, a zoomed view of the Galactic center region. The background maps are the combined exposure maps of *XMM-Newton* EPIC-pn, EPIC-MOS, and *Chandra* ACIS observations (EPIC-MOS and ACIS exposure maps were weighted by 0.4 compared to EPIC-pn exposure maps). *Chandra* detections dominate the central few arcminutes, while *XMM-Newton* detections dominate further away.

with 302 detections (in total), the false alarm rate for our XMM candidate continuum sources is ≈ 0.1.

The CSC v2.0 catalog provides three measures for the probability that the source is variable within an observation— the Gregory-Loredo estimate of variability (`var_prob_h`), the Kolmogorov–Smirnov test for variability (`ks_prob_h`), and Kuiper's test for variability (`kp_prob_h`). We check these values for each "good" observation (check §2.2 for selection criteria) of our candidate sources. For a source to be variable within an observation (short-term variability), we require that at least two of these three values be $>= 0.999$. Given that we have 82 candidate continuum *Chandra* sources with 838 observations (in total), the false alarm probability for our list of *Chandra* sources is ≈ 0.1.

To calculate whether different observations/detections of a given source show inter-observation variability, we compare the flux calculated in each observation in the hard band (2–7 keV for *Chandra*, and 2.0–10.0 keV for *XMM-Newton*) and check the χ^2 probability that the flux is constant. We use a threshold $p < 0.001$ to consider a source to be variable between observations (long-term variability).

Note that showing variability does not exclude an X-ray source from being an MSP. We expect that a majority of the X-ray detectable ($L_X > 10^{32}$ erg/s) Galactic Center MSPs (J. Zhao & C. O. Heinke 2022; J. Berteaud et al. 2021) should be redbacks (including transitional MSPs), which often show strong X-ray variation.

4.3. *Background AGN*

AGN, often at high redshift, make up the majority of X-ray sources in deep X-ray observations at high Galactic latitude. While many AGN show fluorescent Fe lines, their redshift can shift the 6.4 keV line to outside the 6.2-7.2 keV range that we consider here. We use the exposure maps of *XMM Newton* and *Chandra*, the absorption column density, N_H from extinction values of E. F. Schlafly & D. P. Finkbeiner (2011), and the R. Gilli et al. (2007) X-ray background model to estimate an upper limit to the number of AGN among our candidate continuum sources. This is an upper limit estimate, since we consider the extreme case that all AGN lack Fe lines within our selected energy range.

4XMM-DR13 provides separate exposure maps for several energy bands. We notice that the band 4 (2.0–4.5 keV) and band 5 (4.5–12 keV) exposure maps have similar exposure values (exposure values are provided in the units of seconds). Therefore we only combine band 5 exposure maps of all the *XMM Newton* observations we use in our analysis. The exposure maps of different detectors and filters are combined separately (i.e., three combined exposure maps of PN for "thin","medium", and "thick" filters, and three for the different MOS filters). CSC 2.0 provides ACIS exposure maps for three different energy bands. We use the hard band (2–7 keV) exposure maps to calculate total exposures. Since these exposure maps are provided in the units of cm^2 s, we divide the exposure map by 358.13 cm^2 — the effective area of ACIS-I at 3.8 keV, the effective energy of the

Table 3. Descriptive version of the "*XMM-Newton* candidates without lines" table

Number	Column name	Units	Explanation
1	Name	—	IAU name of the XMM source
2	Source ID	—	4XMM-DR13 unique source ID
3	RA	deg	Mean Right ascension in the 4XMM-DR13 `SC_RA` column
4	Dec	deg	Mean Declination in the 4XMM-DR13 `SC_DEC` column
5	Position error	arcsec	Source position error in the 4XMM-DR13 `SC_Poserr` column
6	Variability	—	Flags for short-term (S) and long-term (L) variability
7	Known Source	—	Known counterpart for the source
8	Counterpart	—	Marked as "Y"" if radio counterpart exists, else left blank
9	Ref.	—	Reference for the known source or radio survey with the counterpart

Note—This Table is published in its entirety in the electronic edition of the *Astrophysical Journal*. Here, we only describe the columns for guidance regarding its form and content.

Table 4. Descriptive version of the "*Chandra* candidates without lines" table

Number	Column name	Units	Explanation
1	Name	—	IAU name of the Chandra source
2	RA	deg	Right ascension in the CSC v2.0 master catalog
3	Dec	deg	Declination in the CSC v2.0 master catalog
4	Position error	arcsec	Source position error in the CSC v2.0 master catalog
5	Variability	—	Flags for short-term (S) and long-term (L) variability
6	Known Source	—	Known counterpart for the source
7	Counterpart	—	Marked as "Y"" if radio counterpart exists, else left blank
8	Ref./Comments	—	Reference for the known source or radio survey with the counterpart/other comments.

Note—This Table is published in its entirety in the electronic edition of the *Astrophysical Journal*. Here, we only describe the columns for guidance regarding its form and content.

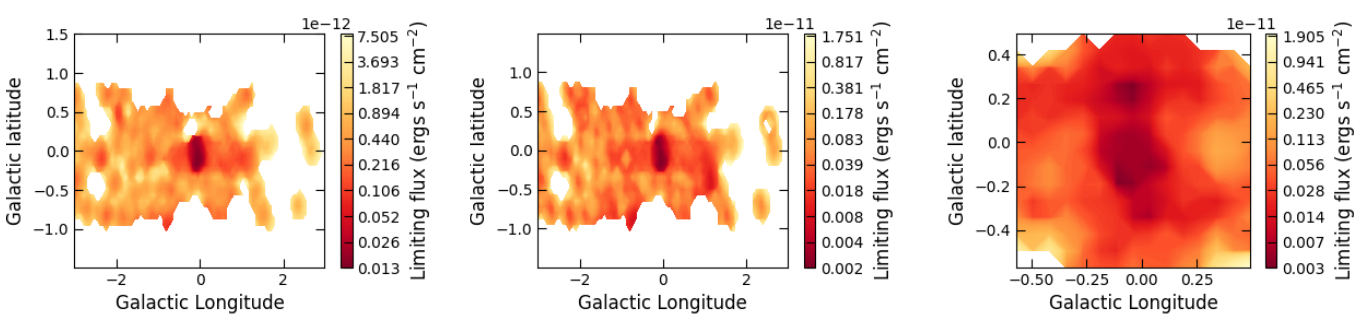

Figure 6. Limiting flux for AGN in our analysis for EPIC-PN (*left*), EPIC-MOS (*center*) & ACIS-I (*right*) detectors. The limiting flux indicates the minimum flux required for the AGN to be detected with $\geq$ 250 counts, after suffering Galactic absorption, given the total exposure time towards that region. The flux is calculated using N_H values and combined exposures of all observations used in our analysis. We use a grid size of $4.5'$.

hard band — to get the value of exposures in seconds (we consider the ACIS-I and ACIS-S exposures together as they have similar responses in the hard band).

We use a grid size of $4.5' \times 4.5'$ for exposure and N_H maps. Since we only consider sources with ≥ 250 counts in the 2–10 keV energy range for identifying candidate continuum sources, we calculate the minimum count rate to reach this threshold for different sky positions based on the exposure values. We then use the corresponding Galactic N_H values and PIMMS v4.13 to calculate the lower limit for the flux of AGN with ≥ 250 counts at that location. We show our resulting limiting flux maps in Fig. 6. We notice that the calculated limiting flux of EPIC-pn observations reaches $\sim 10^{-14}$ ergs/s, but none of the observed XMM sources with counts ≥ 250 had such a low flux. This could be due to the effect of the bright diffuse emission background. Therefore, we place an empirical floor on the limiting flux of 4.0×10^{-14} ergs cm^{-2}/s. We then calculate the number of AGN at each grid point using these values with the R. Gilli et al. (2007) X-ray background model, and estimate the predicted total number of AGN with ≥ 250 counts in our observations. We predict an upper limit of ∼14 AGN in EPIC-pn observations, ∼12 AGN in EPIC-MOS observations and ∼5 AGN in ACIS observations.

The N_H values generated from E. F. Schlafly & D. P. Finkbeiner (2011) maps are significantly higher than the best-fit absorption values for X-ray spectra of our sources. We infer that this is likely due to the majority of our X-ray sources lying near the Galactic Center, while the E. F. Schlafly & D. P. Finkbeiner (2011) maps include dust and gas on the far side of the Galaxy, which we expect would indeed affect distant AGN. Repeating our AGN estimates using N_H values from the best-fitting spectra of our selected X-ray sources (check appendix), we get ~ 50 background AGN among *XMM* detections and ~ 26 AGN among the *Chandra* sources. These numbers would suggest that most of our candidate continuum sources from XMM are background AGN, but we reiterate that we think the true N_H values for background AGN are higher.

4.4. *Probability of CVs among candidate quiescent sources*

While most bright CVs (dominated by intermediate polars [IPs]; V. F. Suleimanov et al. 2022) show strong Fe lines, a few show weaker Fe lines (see e.g. H. Ezuka & M. Ishida 1999). This, along with Poisson noise from low counts in the 5.8-7.6 keV range, and high background diffuse X-ray emission, might make it difficult to detect Fe lines from some CVs, even using spectral analysis. **Therefore, we attempt to simulate CVs similar to those studied in the Galactic Center region. Since most of our candidate sources fall within the region of analysis of S. Mondal et al. (2025) (except for 3 of our sources that have $-0.9 < b < -0.8$), we use their spectral fits of detected and candidate CVs to simulate a sample of CVs in the galactic bulge. S. Mondal et al. (2025) use a power-law fit with Gaussian lines at 6.7 and 7.0 keV for spectral fitting. We use the power-law indices, equivalent width (EW) of the 6.7 keV line (EW$_{6.7}$), and the ratio of line intensities of 7.0 keV & 6.7 keV from S. Mondal et al. (2025), and N_H and L_X values from our MSP simulations, to simulate CVs, and then estimate the number of CVs among our candidate continuum sources. Since S. Mondal et al. (2025) do not report on the strength of the 6.4 keV fluorescent iron line, we use the ratio of intensities of 6.4 keV & 6.7 keV Fe lines reported by X.-j. Xu et al. (2016) for IPs in the solar neighborhood. We also note that this range of EW$_{6.7}$ matches the distribution of EWs in M. P. Muno et al. (2004) in their spectral analyses of individual sources and stacked spectra.**

We compare the simulations in the **top** left panel of Fig. 7 with the observed population of sources with lines in the **top** right panel. The right panels of Fig. 7 **suggest** two populations of observed X-ray sources, one with a distribution of Fe line strength reasonably represented by our CV simulations, and another **(most obvious among the brighter sources, but visible as a distinct population down to ∼200 counts)** with an X-ray color consistent with no Fe line flux, **and inconsistent with our CV simulations** (cf. Fig. 4).

We then select simulated CVs that fall within the **X-ray color** bounds **that** we set for continuum sources **(i.e. candidate MSPs, §2.4 above)**, and perform a spectral analysis of these sources. We check for simulated CVs that do not show evidence of Fe lines by using the same criteria that we use for selecting continuum sources: they do not show significant improvement in the C-statistic when fitting to a model with Fe lines, and/or the photon flux [`norm`] in these lines is consistent with zero. Using calculations detailed in the Appendix, we estimate that there could be ∼**5 CVs among the EPIC-pn detections, ∼3 in EPIC-MOS detections, and ∼11 in ACIS detections, that cannot be distinguished from sources lacking Fe lines**.

The EWs tabulated by S. Mondal et al. (2025) might show a bias towards CVs with strong Fe emission lines, as they only report sources that

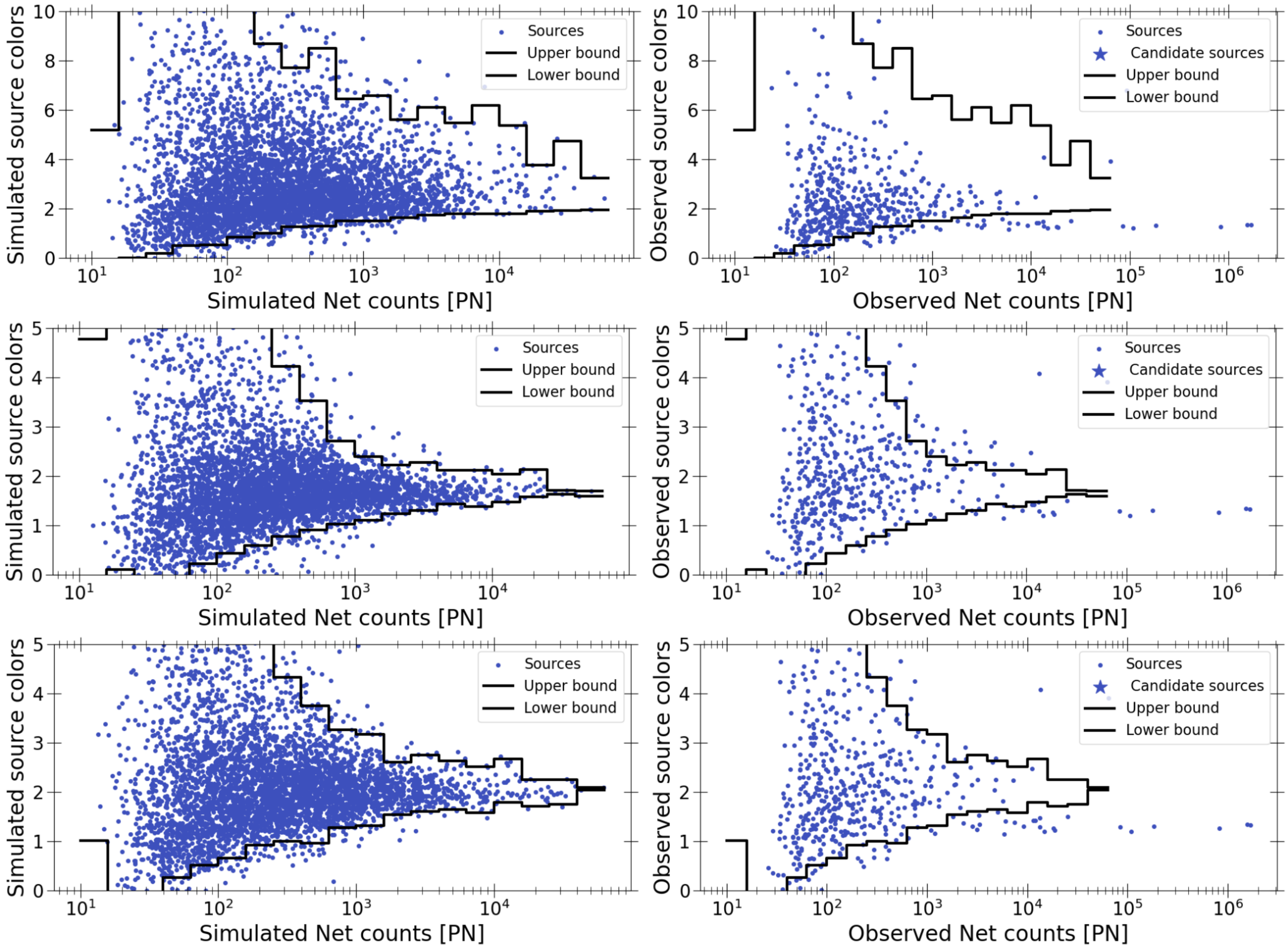

Figure 7. Comparing the X-ray colors of simulated CVs in the EPIC-PN detector with those of the observed PN sources. The left column shows the simulated sources and the right shows the observed PN spectra. The *top* row shows the results of CVs simulated using S. Mondal et al. (2025) EWs, the *middle* row shows results from using the EWs of IPs in X.-j. Xu et al. (2016), and the *bottom* row shows CVs with twice the EW of IPs in X.-j. Xu et al. (2016).

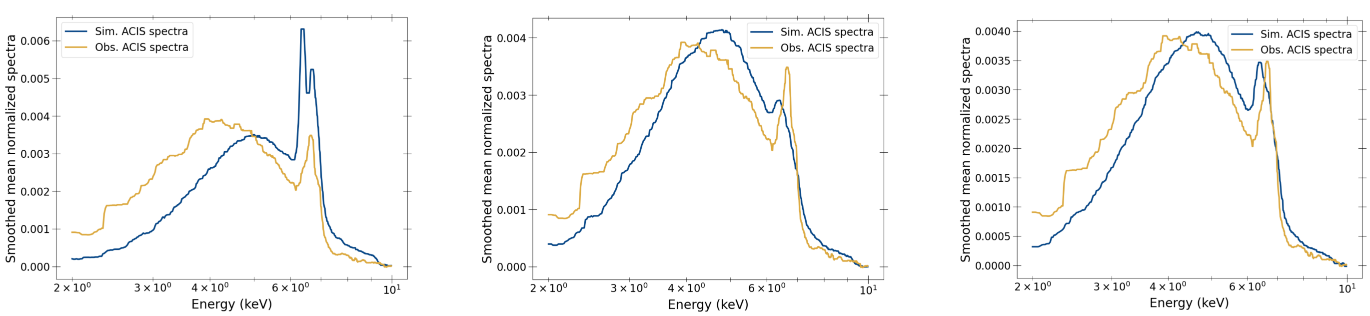

Figure 8. Comparision between the mean normalized ACIS spectrum of the simulated CVs to that of observed sources. The *left* plot shows the simulated CVs using the S. Mondal et al. (2025) EWs; the *middle* plot shows the CVs simulated using EW of IPs in X.-j. Xu et al. (2016); and the right plot shows CVs simulated with twice the EW of IPS in X.-j. Xu et al. (2016).

show the presence of a 6.7 keV Fe line. It is possible that there are additional CVs that for some reason (e.g. low metallicity) do not show significant Fe lines, contaminating our sample of apparently line-less X-ray sources. To get an upper limit of the number of CVs among our candidate sources, we use the X.-j. Xu et al. (2016) EW values for IPs in the solar neighborhood. Our limiting luminosity for sources to have >250 counts is $\gtrsim 10^{32}$ ergs/s for *Chandra* data, or $\gtrsim 10^{33}$ ergs/s for *XMM* PN data. Very few dwarf novae reach a maximum X-ray luminosity above 10^{32} erg/s (A. D. Schwope et al. 2024 list 131 dwarf novae with eROSITA fluxes and Gaia parallaxes, of which the 4 highest log L_X values are 32.6, 32.5, 32.1, 32.1). Any CVs among our candidates are probably dominated by IPs. IPs have the smallest average EWs among all the different types of CVs analyzed in X.-j. Xu et al. (2016)), so this is the most conservative choice. We use the plasma temperature values from X.-j. Xu et al. (2016) in our simulations. Since X.-j. Xu et al. (2016) use zero metallicity for the `apec` component (modeling the lines separately), we set the `apecnolines` flag to `yes`; i.e., all the Fe emission is modelled by three Gaussian lines at 6.4 keV, 6.7 keV and 7.0 keV. Repeating our calculations on these simulations gives us ∼24 CVs among our PN candidate continuum sources, ∼16 in our MOS candidates, and ∼56 in our ACIS candidates. I.e., if CVs in the Galactic Center were well-represented by the spectra of local IPs as measured by X.-j. Xu et al. (2016), then half or more of our candidate continuum sources would likely be CVs with weak Fe lines.

However, comparison of the distribution of X-ray colors for these simulated CVs with the observed Galactic Bulge sources (middle plots of Fig. 7) shows that the local CVs studied by X.-j. Xu et al. (2016) have significantly lower X-ray colors. Quantitatively, the average X-ray colors (defined in §2.4; given for pn/MOS/ACIS) for simulated CVs with $2 * 10^3 - 10^4$ counts, and excluding sources with X-ray colors consistent (within 2σ) with continuum-only sources, are 1.68/1.58/1.62, while the observed sources with the same restrictions have average X-ray colors of 2.0/2.2/2.1. This shows that the distribution of X-ray color among CVs in the Galactic Bulge is inconsistent with those in local space reported by X.-j. Xu et al. (2016). This is consistent with the literature; the 6.7 keV Fe line EWs of IPs reported in X.-j. Xu et al. (2016) are much smaller (average 107 eV) than those reported for Galactic Center sources of similar L_X by M. P. Muno et al. (2004) (350 eV), X.-j. Xu et al. (2019) (269 eV), and S. Mondal et al. (2025) (415 eV), and also much smaller than the EW of Fe lines in Galactic Ridge X-ray emission as reported in X.-j. Xu et al. (2016) (490 eV). This indicates that modeling the CVs in the Galactic Bulge using Fe line strengths from X.-j. Xu et al. (2016) is rather too conservative, and would suggest more weak-line CVs than actually exist among our candidate sources.

As another alternative, we shift the X-ray color distribution of X.-j. Xu et al. (2016), increasing it by a factor of two. This is motivated by the supersolar metallicity (factor 1.9) of the Galactic Nuclear Disk region that we study, derived by K. Anastasopoulou et al. (2023) from X-ray line emission. (Measurements of the metallicity difference betweeen the Galactic Nuclear Disk, e.g. M. Schultheis et al. 2021, vs. local space, e.g. L. Casagrande et al. 2011, suggest a smaller difference of $\lesssim 1.6$, but the nuclear disk metallicity is still somewhat uncertain.) CVs simulated from twice the EWs of Xu et al. IPs seem to match the observed Fe line strengths more accurately (Fig. 7). Comparing the double-Xu vs. Mondal distributions to the observed sources using the average X-ray color test above, we find that the Mondal distribution gives average X-ray colors (pn/MOS/ACIS, again excluding sources consistent with continuum sources) of 3.01/2.80/2.87, while the double-Xu simulation gives X-ray colors of 2.03/1.92/1.95, very close to the average observed X-ray colors of 2.0/2.1/2.2. We illustrate the match of the twice-Xu simulations with the observed data from all three detectors in Fig. 9.

As a final comparison, we plot the mean normalized ACIS spectrum of the simulated CVs, using the three different tested weighting schemes, to those of the observed sources in Fig. 8. The average observed ACIS spectrum seems most similar to the twice-Xu metallicity simulations, though we note that these average spectra include continuum sources.

Using the simulated sources assuming double the Xu et al. Fe line strengths, we estimate ∼11 CVs among PN candidate continuum sources, ∼4 CVs among MOS candidates and ∼16 CVs among the ACIS candidates.

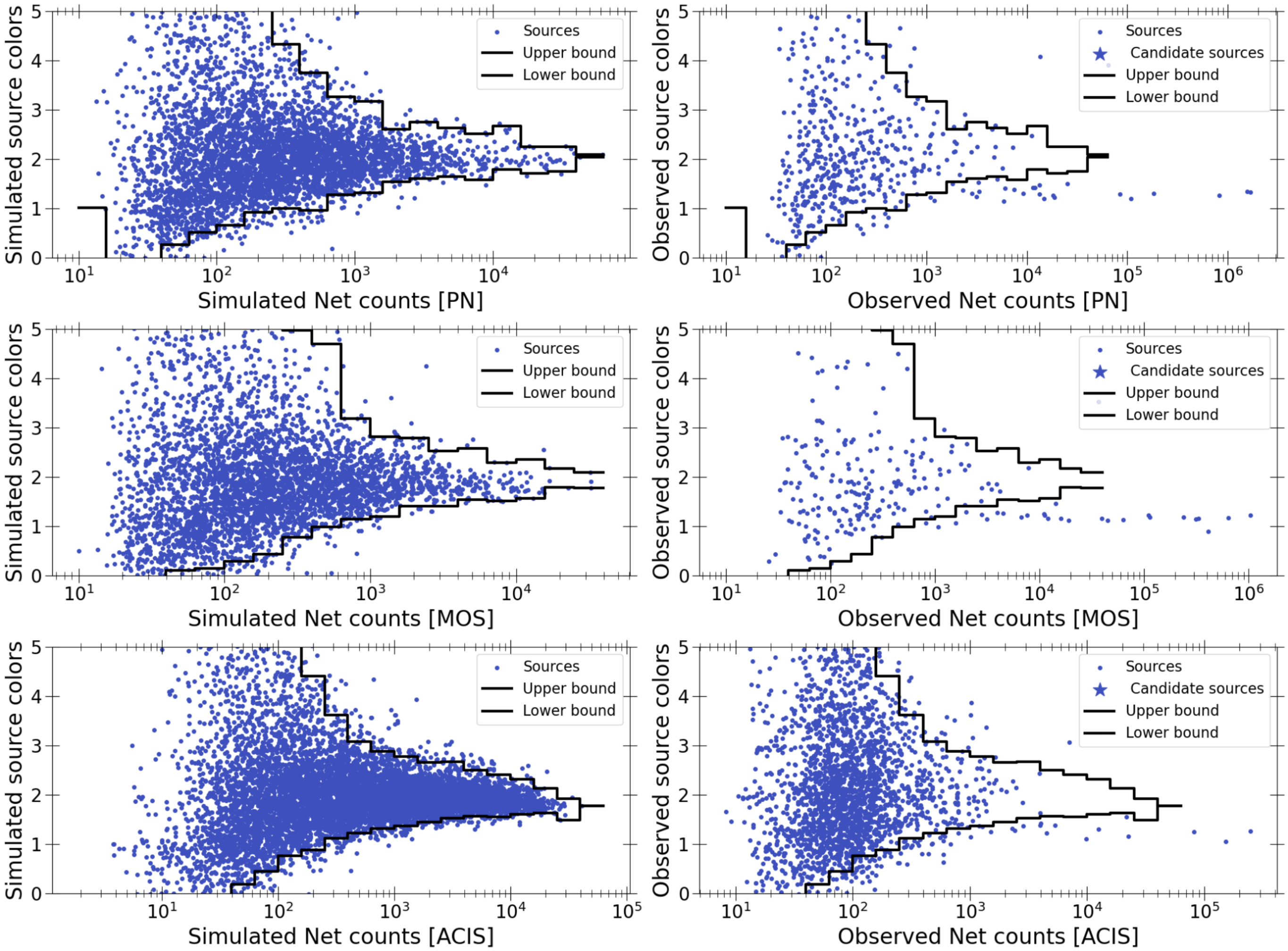

Figure 9. Comparing the X-ray colors of simulated CVs (using twice the EWs of IPs from X.-j. Xu et al. 2016) in each detector, with those of the observed sources. The left column shows the simulated CVs, and 95% upper and lower bounds, while the right shows the observed spectra. The *top* row shows the results for the XMM-Newton pn detector, the *middle* row shows results for the XMM-Newton MOS detector, and the *bottom* row shows results for the *Chandra* ACIS detector.

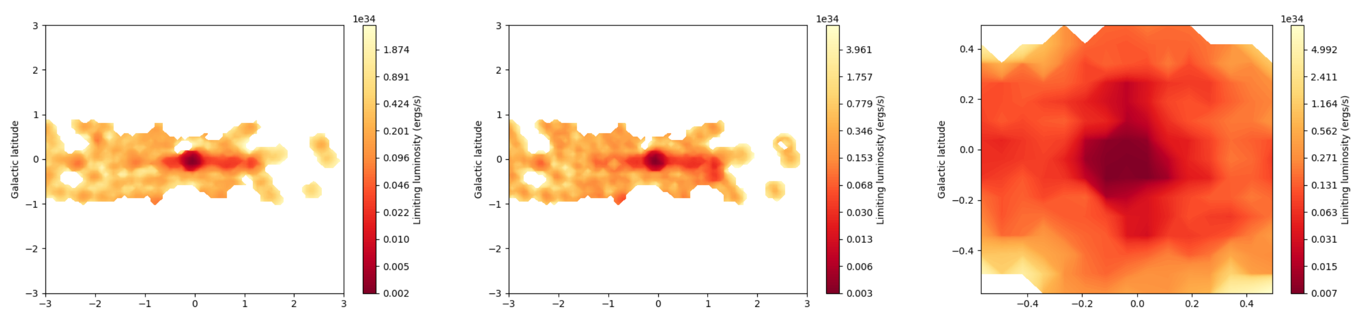

Figure 10. Limiting unabsorbed luminosity for GB sources to have ≥250 counts in our analysis for EPIC-PN (*left*), EPIC-MOS (*center*) & ACIS-I (*right*) detectors. The flux is calculated using N_H values and combined exposures of all observations used in our analysis. We use a grid size of $4.5'$.

4.5. *Implications for the γ-ray GCE*

We use the M. Di Mauro (2021) estimates of γ-ray flux around the galactic center, the γ-ray luminosity function of MSPs (P. L. Gonthier et al. 2018; D. Hooper & G. Mohlabeng 2016), and the X-ray luminosity function of MSPs in globular clusters (J. Zhao & C. O. Heinke 2022) to compare the numbers of candidate continuum sources we detect to the predicted number of detectable MSPs, if the γ-ray GCE is indeed produced by MSPs. First, we compare the number of *Chandra* candidates in the central 0.2° around Sgr A*. We fit the $dS/d\Omega$ vs θ values in Fig. 10 of M. Di Mauro (2021) (where $dS/d\Omega$ is the γ-ray surface brightness calculated with annuli of size 1°, and θ is the angular distance from the Galactic Center) with a power-law (i.e, $dS/d\Omega = A(\theta/1^\circ)^{-\Gamma}$, where Γ is the power-law index). The best-fit power-law index is 1.55 ± 0.05 and power-law norm is 0.047 ± 0.004. Using this power-law fit, we estimate the γ-ray flux within 0.1° and 0.2° to be $(1.2 \pm 0.4) \times 10^{-10}$ ergs/cm^2/s and $(1.6 \pm 0.5) \times 10^{-10}$ ergs/cm^2/s, respectively, corresponding to a GCE luminosity of $(9 \pm 3) \times 10^{35}$ ergs/s and $(1.3 \pm 0.4) \times 10^{36}$ ergs/s, for a Galactic Center distance of 8.0 kpc.

We use the various spectral energy distributions of MSPs presented in P. L. Gonthier et al. (2018) to convert these 1–10 GeV luminosities to 0.1–1000 GeV luminosity, L_γ. Depending on the SED used, we get L_γ values of $(1.8 \pm 0.7) \times 10^{36}$ ergs/s within 0.1° and $(2.5 \pm 0.9) \times 10^{36}$ ergs/s within 0.2°. For an average MSP γ-ray luminosity of $< L_{\gamma,MSP} >= 1.1 \times 10^{33}$ ergs/s (GCE MSPs from P. L. Gonthier et al. 2018), this gives ≈ 1700 MSPs within 0.1° and ≈ 2300 MSPs within 0.2°. Alternatively, D. Hooper & G. Mohlabeng (2016) gives $< L_{\gamma,MSP} >= 3.0 \times 10^{33}$ ergs/s for MSPs in globular clusters. If we use this value for MSPs in the GB, we get ≈ 600 and ≈ 800 MSPs within 0.1° and 0.2°, respectively. Another way of estimating the number of MSPs is to consider the fraction of 1-10 GeV flux within 0.1° and 0.2° compared to the 7° (region of interest in P. L. Gonthier et al. 2018). Our fit to the M. Di Mauro (2021) GCE surface brightness gives a flux of $(7.8 \pm 0.8) \times 10^{-10}$ ergs/cm^2/s within 7°. For 11,500 MSPs within 7° (the estimate of P. L. Gonthier et al. 2018), this gives ≈ 1800 and ≈ 2400 MSPs within 0.1 and 0.2° of the galactic center, which is similar to our first estimate.

We correct the N_H values for GB sources (see Appendix A) and calculate the limiting luminosity for GB X-ray sources that can be detected with ≥ 250 counts in the 2–10 keV energy range by the EPIC-pn, EPIC-MOS and ACIS detectors, given the exposure times **(Fig. 10)**. Our 2–10 keV limiting luminosity reaches $\approx 10^{32}$ ergs/s around Sgr A* in the ACIS observations. From the J. Zhao & C. O. Heinke (2022) luminosity function, we notice that only 6 of the 173 MSPs have $L_{2-10keV} > 10^{32}$ ergs/s. Thus, we should only be able to detect 20–62 and 28–83 MSPs within 0.1° and 0.2°. Our analysis of *Chandra* sources has revealed **83** candidate continuum sources (mostly within 0.2°), of which we have estimated ~ 5 possible AGN and ~ 16 possible CVs. If **many** of the remaining line-less X-ray sources are MSPs, they could be enough to produce the observed GCE.

At an angular distance of 1.5° away from the galactic center, the γ-ray GCE surface-brightness reduces to 3×10^{-8} ergs/cm^2/s/sr (which translates to ~ 100 MSPs per square degree) and falls off exponentially. However, our limiting X-ray luminosity in these regions is $\gtrsim 10^{33}$ ergs/s, i.e. approximately 1 out of every 173 MSPs will be detected. Thus we expect very few MSPs outside the central region; with about 30 XMM fields around this radius, each of ∼0.2 square degrees, this suggests of order 4 detectable MSPs. This is consistent with our previous observation that the number of candidate line-less X-ray sources in the XMM observations is similar to the sum of CVs with **weak** Fe emission, plus background AGN, **and other continuum sources**.

5. SUMMARY & CONCLUSION

In this paper, we attempted to identify MSPs in the GB to constrain the gamma-ray excess observed in Fermi observations of the GC. We searched for *Chandra* and *XMM-Newton* sources in the GB that exhibit continuum spectra and do not show fluorescent Fe or Fe-K lines typical of CVs. We find a total of 54 *XMM-Newton* sources and 83 *Chandra* sources, among which 16 X-ray sources are known X-ray sources including known X-ray binaries, one PWN and one magnetar. We estimate that ∼25 of the *XMM-Newton* sources and ∼5 *Chandra* sources could be background AGN. Our candidate sources are also likely to include 10–20 CVs with weak Fe lines **from each instrument, as well as young pulsars, X-ray binaries, etc.** Comparison of the number of lineless XMM candidates with the number of possible CVs and background AGN show that only a small number **($\lesssim 30$)** of XMM candidates **may be** MSPs. Calculating the number of MSPs needed to explain the GCE surface brightness distribution shows that most of the detectable MSPs will be close to the Galactic Center, and could be detected in our ACIS list of continuum X-ray sources. Indeed, our list of 83 *Chandra* candidate continuum sources **(likely ∼60, when removing AGN and weak-line CVs),** is consistent with the expected number of 20-80 MSPs detectable in the X-ray, if the γ-ray GCE is indeed produced by MSPs.

The high absorption from dust and gas towards the Galactic center and the large density of stars in the Galactic plane give us limited information on the counterparts for our candidate sources. Nevertheless, 16 of our suggested candidates have been independently identified as X-ray binaries or pulsars/magnetars/pulsar wind nebulae. We suggest that further follow-up of unclassified radio sources that have potential matches with these X-ray sources may be particularly rewarding.

The bright diffuse emission close to the Galactic center increases the background noise in the detected X-ray sources. Future X-ray telescopes with high effective area and a narrow point-spread-function (e.g. the proposed AXIS telescope; C. S. Reynolds et al. 2023; S. Safi-Harb et al. 2023) would allow us to reach lower limiting luminosities, and constrain the population of MSPs.

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APPENDIX

A. CALCULATING ABSORPTION COLUMN FOR GALACTIC BULGE SOURCES.

The N_H values calculated from the dust extinction maps of E. F. Schlafly & D. P. Finkbeiner (2011) will include the Galactic absorption throughout the Galaxy. Therefore, they overestimate the absorption column for sources detected within the Galactic Bulge (GB). We need to calculate the absorption column for GB sources to determine the limiting luminosity of MSPs detectable in our observations and estimate the number of detectable MSPs required to explain the Fermi GCE.

We utilized our spectral analysis of the 116 *XMM-Newton* sources and 130 *Chandra* sources to map the N_H in the GB. We compare the best-fitting N_H value from spectra of sources ($N_{H,spec}$) with that calculated from dust extinction ($N_{H,dust}$) around the sources (Fig. 11). From this comparison, we approximate:

$$N_{H,spec} = N_{H,dust}, \text{ if } N_{H,dust} < 10^{23} \text{ cm}^2, \tag{A1}$$

$$N_{H,spec} = 0.5 N_{H,dust}, \text{ if } N_{H,dust} \in (10^{23}, 5 \times 10^{23}) \text{ cm}^2, \tag{A2}$$

$$N_{H,spec} = 0.2 N_{H,dust}, \text{ if } N_{H,dust} > 5 \times 10^{23} \text{ cm}^2 \tag{A3}$$

The smaller values of $N_{H,spec}$ w.r.t $N_{H,dust}$ are consistent with our line of sight passing through larger amounts of gas beyond the GB as we approach the Galactic Center.

B. ESTIMATING THE NUMBER OF CATACLYSMIC VARIABLES IN OUR SAMPLE.

For our purposes, we assume that all sources considered are either continuum sources (CS) or cataclysmic variables (CV). For calculations below, we only consider sources whose Fe colors are below the 2σ upper limit of continuum colors. We denote sources that do not reveal Fe lines upon spectroscopic analysis by NL (no lines). To estimate the number of CVs among our candidate sources, we first calculate the probability that an observed source with no detected Fe lines is a CV, i.e., $p(CV|NL)$.

From Bayes' theorem, $p(CV|NL) = p(NL|CV) * p(CV/p(NL)$; where $p(CV)$ is the probability that the observed source is a CV, and $p(NL)$ is the probability that the observed source has no Fe lines detected in its spectra. Since we assume all our sources are either continuum or cataclysmic variables, we can write,

$$p(NL) = p(NL|CS)p(CS) + p(NL|CV)p(CV) = p(NL|CS)(1 - p(CV)) + p(NL|CV)p(CV), \tag{B4}$$

$$\Rightarrow p(CV) = \frac{p(NL|CS) - p(NL)}{p(NL|CS) - p(NL|CV)}, \tag{B5}$$

where $p(CV)$, $p(CS)$ are the probabilities that the observed source is a CV, or a continuum source; $p(NL|CS)$, $p(NL|CV)$ are the probabilities that no Fe lines are detected in the X-ray spectra of a continuum source, or a CV.

We can estimate $p(NL|CV)$ and $p(NL|CS)$ from our simulated samples of Galactic CVs and MSPs, and approximate $p(NL)$ from the observed sample of sources. We divide sources with net counts $\geq$ 250 into 5 bins and calculate

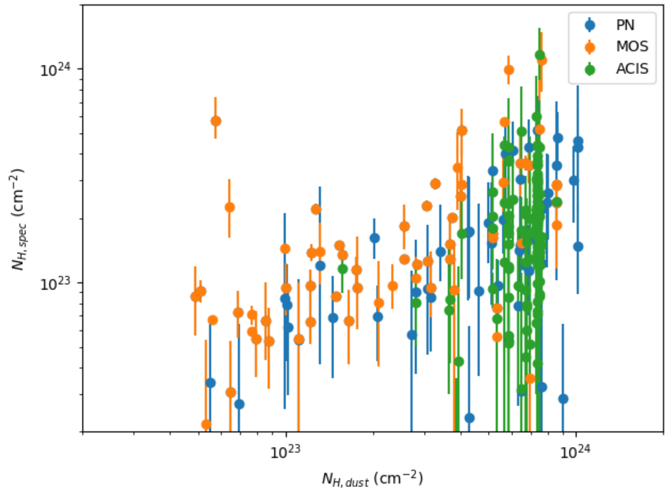

Figure 11. Comparing the absorption column density from spectral fitting ($N_{H,spec}$) to that calculated from dust extinction maps ($N_{H,dust}$).

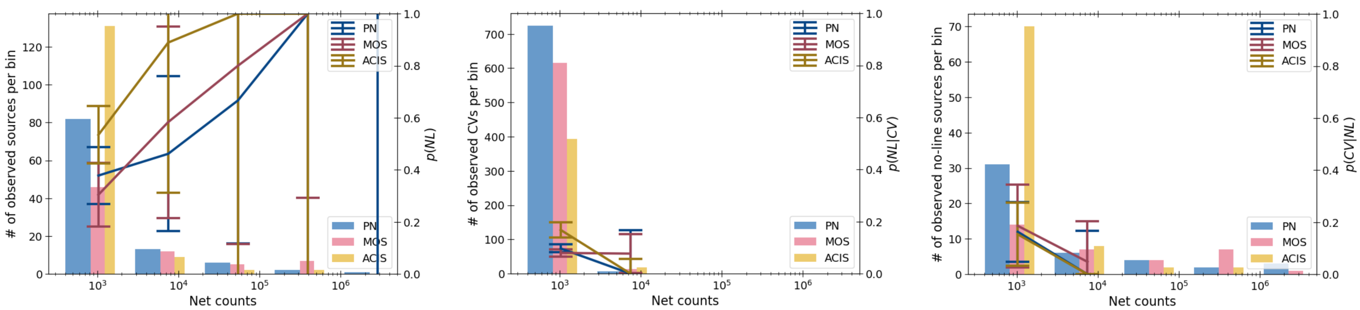

Figure 12. *Left*: Probabilities that observed sources have no detected Fe lines $p(NL)$; *Center*: probabilities that simulated CVs do not show sufficient evidence of Fe lines i.e. $p(Nl|CV)$; *Right*: probability that sources with no detected Fe lines could be CVs. All results shown here are for sources that have Fe colors consistent with that of MSPs. We also show the histograms of the observed sources, simulated CVs, and candidate continuum sources (with MSP-like colors). Given that none of our simulated CVs with ≥ 3000 counts had colors consistent with simulated MSPs, we assume that all sources with ≥ 3000 counts and showing no lines on spectral analysis are continuum sources.

$p(CV|NL)$ for each bin. Since the number of observed candidate continuum sources (NL) with $\geq$ 1500 counts is limited, the error-bars on $p(NL)$ in these bins can be large. Therefore, in bins where $p(NL) > p(NL|CS)$, we use $p(CV) = 0$. We then multiply this by the number of candidate sources in each bin to estimate the number of CVs in our sample.

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